

8

The mean free path and collisions

8.1 The mean collision time	68
8.2 The collision cross-section	69
8.3 The mean free path	71
Chapter summary	72
Exercises	72

¹It turns out that large-angle scattering dominates transport processes in most gases (described in Chapter 9) and is largely independent of energy and therefore temperature; this allows us to use a rigid-sphere model of collisions, i.e. to model atoms in a gas as billiard balls.

At room temperature, the r.m.s. speed of O_2 or N_2 is about 500 ms^{-1} . Processes such as the diffusion of one gas into another would therefore be almost instantaneous, were it not for the occurrence of *collisions* between molecules. Collisions are fundamentally quantum mechanical events, but in a dilute gas, molecules spend most of their time between collisions and so we can consider them as classical billiard balls and ignore the details of what actually happens during a collision. All that we care about is that after collisions the molecules' velocities become essentially randomized.¹ In this chapter we will model the effect of collisions in a gas and develop the concepts of a mean collision time, the collision cross-section and the mean free path.

8.1 The mean collision time

In this section, we aim to calculate the average time between molecular collisions. Let us consider a particular molecule moving in a gas of other similar molecules. To make things simple to start with, we suppose the molecule under consideration is travelling at speed v and that the other molecules in the gas are stationary. This is clearly a gross oversimplification, but we will relax this assumption later. We will also attribute a collision cross-section σ to each molecule which is something like the cross-sectional area of our molecule. Again, we will refine this definition later in the chapter.

In a time dt , our molecule will sweep out a volume $\sigma v dt$. If another molecule happens to lie inside this volume, there will be a collision. With n molecules per unit volume, the probability of a collision in time dt is therefore $n\sigma v dt$. Let us define $P(t)$ as follows:

$$P(t) = \text{the probability of a molecule not colliding up to time } t. \quad (8.1)$$

Elementary calculus then implies that

$$P(t + dt) = P(t) + \frac{dP}{dt} dt, \quad (8.2)$$

but $P(t + dt)$ is also the probability of a molecule not colliding up to time t multiplied by the probability of not colliding in subsequent time dt , i.e. that

$$P(t + dt) = P(t)(1 - n\sigma v dt). \quad (8.3)$$

Hence rearranging gives

$$\frac{1}{P} \frac{dP}{dt} = -n\sigma v \quad (8.4)$$

and therefore that (using $P(0) = 1$)

$$P(t) = e^{-n\sigma vt}. \quad (8.5)$$

Now the probability of surviving without collision up to time t but then colliding in the next dt is

$$e^{-n\sigma vt} n\sigma v dt. \quad (8.6)$$

We can check that this is a proper probability by integrating it,

$$\int_0^\infty e^{-n\sigma vt} n\sigma v dt = 1, \quad (8.7)$$

and confirming that it is equal to unity. Here, use has been made of the integral

$$\int_0^\infty e^{-x} dx = 0! = 1 \quad (8.8)$$

(see Appendix C.1). We are now in a position to calculate the **mean scattering time** τ , which is the average time elapsed between collisions for a given molecule. This is given by

$$\begin{aligned} \tau &= \int_0^\infty t e^{-n\sigma vt} n\sigma v dt \\ &= \frac{1}{n\sigma v} \int_0^\infty (n\sigma vt) e^{-n\sigma vt} d(n\sigma vt) \\ &= \frac{1}{n\sigma v} \int_0^\infty x e^{-x} dx \end{aligned} \quad (8.9)$$

where the integral has been simplified by the substitution $x = n\sigma vt$. Hence we find that

$$\tau = \frac{1}{n\sigma v}, \quad (8.10)$$

where use has been made of the integral (again, see Appendix C.1)

$$\int_0^\infty x e^{-x} dx = 1! = 1. \quad (8.11)$$

8.2 The collision cross-section

In this section we will consider the factor σ in much more detail. To be as general as possible, we will consider two spherical molecules of radii a_1 and a_2 with a **hard-sphere potential** between them (see Fig. 8.1).

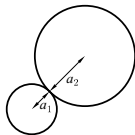


Fig. 8.1 Two spherical molecules of radii a_1 and a_2 with a hard-sphere potential between them.

This implies that there is a potential energy function $V(R)$ that depends on the relative separation R of their centres, and is given by

$$V(R) = \begin{cases} 0 & R > a_1 + a_2 \\ \infty & R \leq a_1 + a_2 \end{cases} \quad (8.12)$$

and this is sketched in Fig. 8.2.

The **impact parameter** b between two moving molecules is defined as the distance of closest approach that would result if the molecular trajectories were undeflected by the collision. Thus for a hard-sphere potential there is only a collision if the impact parameter $b < a_1 + a_2$. Focus on one of these molecules (let's say the one with radius a_1). This is depicted in Fig. 8.3. Now imagine molecules of the other type (with radius a_2) nearby. A collision will only take place if the centre of these other molecules comes inside a tube of radius $a_1 + a_2$ (so that the molecule labelled A would not collide, whereas B and C would). Thus our first molecule can be considered to sweep out an imaginary tube of space of cross-sectional area $\pi(a_1 + a_2)^2$ that defines its 'personal space'. The area of this tube is called the **collision cross-section** σ and is then given by

$$\sigma = \pi(a_1 + a_2)^2. \quad (8.13)$$

If $a_1 = a_2 = a$, then

$$\sigma = \pi d^2 \quad (8.14)$$

where $d = 2a$ is the molecular diameter.

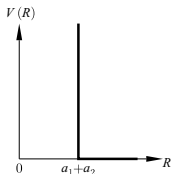


Fig. 8.2 The hard-sphere potential $V(R)$.

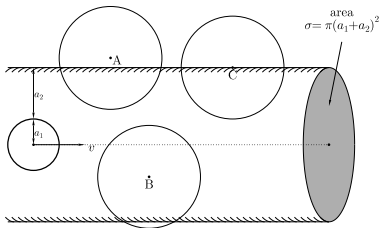


Fig. 8.3 A molecule sweeps out an imaginary tube of space of cross-sectional area $\sigma = \pi(a_1 + a_2)^2$. If the centre of another molecule enters this tube, there will be a collision.

²But not too low a temperature, or quantum effects become important.

³Cross-sections in nuclear and particle physics can be much larger than the size of the object, expressing the fact that an object (in this case a particle) can react strongly with things a long distance away from it.

Is the hard-sphere potential correct? It is a good approximation at lower temperatures,² but progressively worsens as the temperature increases. Molecules are not really hard spheres but slightly squashy objects, and when they move at higher speeds and plough into each other with more momentum, you need more of a direct hit to cause a collision. Thus as the gas is warmed, the molecules may appear to have a smaller cross-sectional area.³

8.3 The mean free path

Having derived the mean collision time, it is tempting to derive the **mean free path** as

$$\lambda = \langle v \rangle \tau = \frac{\langle v \rangle}{n\sigma v} \quad (8.15)$$

but what should we take as v ? A first guess is to use $\langle v \rangle$, but that turns out to be not quite right. What has gone wrong?

Our picture of molecular scattering has been to focus on one molecule as the moving one, and think of all of the others as sitting ducks, fixed in space waiting patiently for a collision to occur. The reality is quite different: all molecules are whizzing around. We should therefore take v as the average relative velocity, i.e. $\langle v_r \rangle$, where

$$\mathbf{v}_r = \mathbf{v}_1 - \mathbf{v}_2 \quad (8.16)$$

and $\mathbf{v}_1$ and $\mathbf{v}_2$ are the velocities of two molecules labelled 1 and 2. Now,

$$v_r^2 = v_1^2 + v_2^2 - 2\mathbf{v}_1 \cdot \mathbf{v}_2, \quad (8.17)$$

so that

$$\langle v_r^2 \rangle = \langle v_1^2 \rangle + \langle v_2^2 \rangle = 2\langle v^2 \rangle, \quad (8.18)$$

because $\langle \mathbf{v}_1 \cdot \mathbf{v}_2 \rangle = 0$ (which follows because $\langle \cos \theta \rangle = 0$). The quantity which we want is $\langle v_r \rangle$, but what we have an expression for is $\langle v_r^2 \rangle$. If the probability distribution is a Maxwell-Boltzmann distribution, then the error in writing $\langle v_r \rangle \approx \sqrt{\langle v_r^2 \rangle}$ is small,⁴ so to a reasonable degree of approximation we can write

$$\langle v_r \rangle \approx \sqrt{\langle v_r^2 \rangle} \approx \sqrt{2} \langle v \rangle \quad (8.19)$$

and hence we obtain an expression for λ as follows:

$$\lambda \approx \frac{1}{\sqrt{2}n\sigma}. \quad (8.20)$$

Substitution of $p = nk_B T$ yields the expression

$$\lambda \approx \frac{k_B T}{\sqrt{2}p\sigma}. \quad (8.21)$$

To increase the mean free path by a certain factor, the pressure needs to be decreased by the same factor.

⁴Equation 7.23 implies that $\langle v \rangle / \sqrt{\langle v^2 \rangle} = \sqrt{\frac{8}{3\pi}} = 0.92$, so the error is less than 10%.

Example 8.1

Calculate the mean free path for a gas of N_2 at room temperature and pressure. (For N_2 , take $d = 0.37$ nm.)

Solution:

The collision cross-section is $\pi d^2 = 4.3 \times 10^{-19} \text{ m}^2$. We have $p \approx 10^5 \text{ Pa}$ and $T \approx 300 \text{ K}$, so the number density is $n = p/k_B T \approx 10^5 / (1.38 \times 10^{-23} \times 300) \approx 2 \times 10^{25} \text{ m}^{-3}$. This leads to $\lambda \approx 1/(\sqrt{2}n\sigma) = 6.8 \times 10^{-8} \text{ m}$.

Notice that both λ and τ decrease with increasing pressure at fixed temperature. Thus the frequency of collisions *increases* with increasing pressure.

Chapter summary

- The mean scattering time is given by

$$\tau = \frac{1}{n\sigma\langle v_r \rangle},$$

where the collision cross-section is $\sigma = \pi d^2$, d is the molecular diameter and $\langle v_r \rangle \approx \sqrt{2}\langle v \rangle$.

- The mean free path is

$$\lambda \approx \frac{1}{\sqrt{2}n\sigma}.$$

Exercises

- (8.1) What is the mean free path of an N_2 molecule in an ultra-high-vacuum chamber at a pressure of 10^{-10} mbar? What is the mean collision time? The chamber has a diameter of 0.5 m. On average, how many collisions will the molecule make with the chamber walls compared with collisions with other molecules? If the pressure is suddenly raised to 10^{-6} mbar, how do these results change?
- (8.2) (a) Show that the root mean square free path is given by $\sqrt{2}\lambda$ where λ is the mean free path.
 (b) What is the most probable free path length?
 (c) What percentage of molecules travel a distance greater (i) than λ , (ii) than 2λ , (iii) than 5λ ?
- (8.3) Show that particles hitting a plane boundary have travelled a distance $2\lambda/3$ perpendicular to the plane since their last collision, on average.
- (8.4) A diffuse cloud of neutral hydrogen atoms in space has a temperature of 50 K. Estimate the mean scattering time (in years) between hydrogen atoms in the cloud and the mean free path (in Astronomical Units). (1 Astronomical Unit is the Earth-Sun distance; see Appendix A for a numerical value.)